

Kooperatives Fliegen

Consensus

Summer Camp 2026
Babak Salamat

Agenda

Multi-agent systems and cooperative control

Graph theory: direction, connectivity, spanning trees, balance

Continuous-time consensus: protocol, value, convergence

Discrete-time consensus: Perron matrix and step size

Performance and failure modes: spectrum to behavior

Python lab bridge: 25 progressive examples

Introduction

Overview of Multi-Agent UAV Systems

- Multi-Agent Systems (MAS) consist of multiple agents that work together to achieve common goals.
- In UAV systems, MAS enables UAVs to perform tasks such as exploration, surveillance, and rescue in a collaborative way.
- MAS are inspired by collaborative behaviors observed in nature, like bird migration and fish schooling, providing a foundation for cooperative control strategies.
- Key Applications: Robotics (drone swarms), transportation (autonomous cars), and communication networks (smart grids).

A mechanical realization of the Frenkel–Kontorova (FK) model

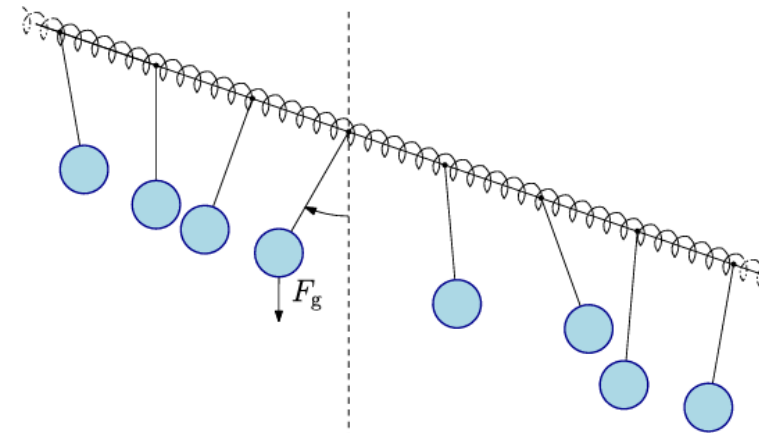


Fig. 1. A series of coupled pendulums pivoting around a single axis.

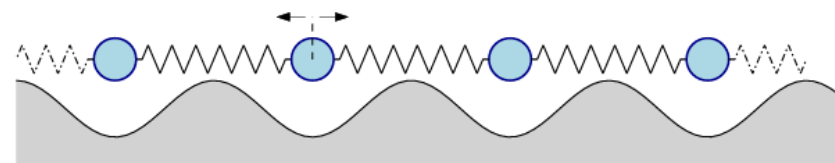
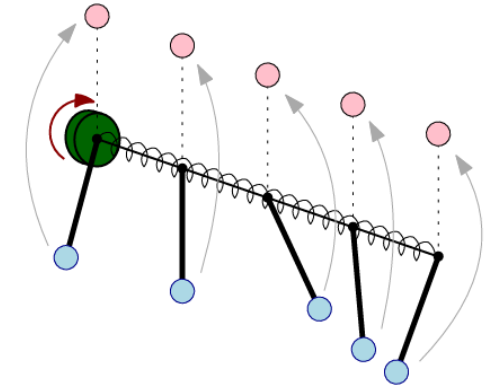


Fig. 2. The FK model as an array of particles in a potential field.



Ref. Loi, Do et al. "Controlled synchronization of coupled pendulums by Koopman Model Predictive Control." *Control Engineering Practice* (2022).

Synchronization extends the life time of the desired behavior of globally coupled systems

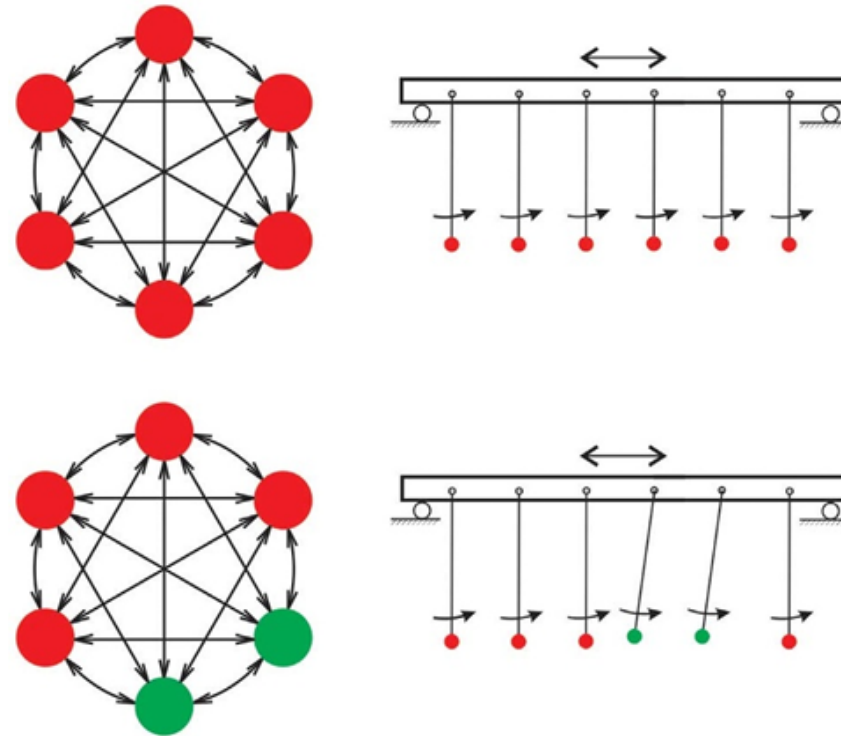
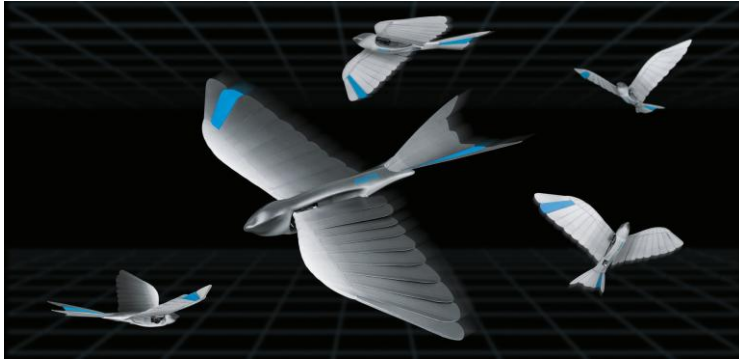


Figure 1 | System of 6 globally coupled units and its mechanical implementation as a system of 6 externally forced pendula mounted to the beam which can move horizontally, (a) all units (pendula) are externally excited (red color) and synchronized, (b) two units (green color) are not excited – the cluster of four excited pendula is phase synchronized with the cluster of two unexcited pendula.

Ref. Synchronization extends the life time of the desired behavior of globally coupled systems M Kapitaniak, M Lazarek, M Nielacny, K Czołczynski, P Perlikowski, Scientific Reports 4 (1), 4391

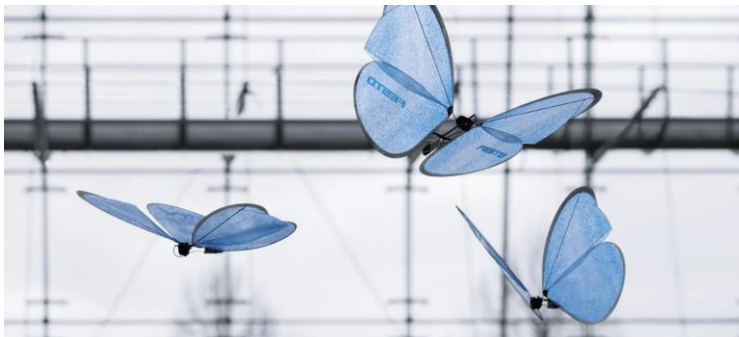
Gleichzeitige Bewegung biologischer Gruppen

Einige Beispiele



Safe aerial acrobatics as a swarm

Link: <https://www.festo.com>



Birds flock

Link: <https://unsplash.com/de/s/fotos/flock-of-birds>

eMotionButterflies with collective behaviour

Link: <https://www.festo.com>

Schwarm in der Natur

Verschiedene Graphtopologien

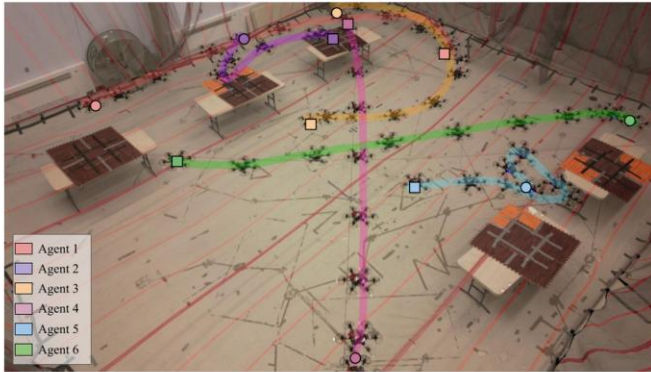


- Behavior on Different Graph Topologies

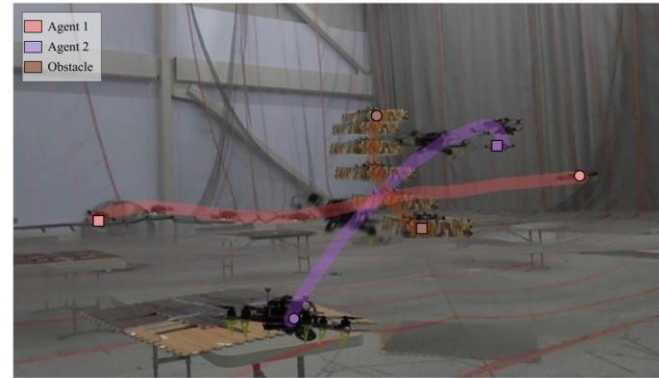
- Birds communicate in ways that differ from fish
- Therefore, their dynamical motion is different



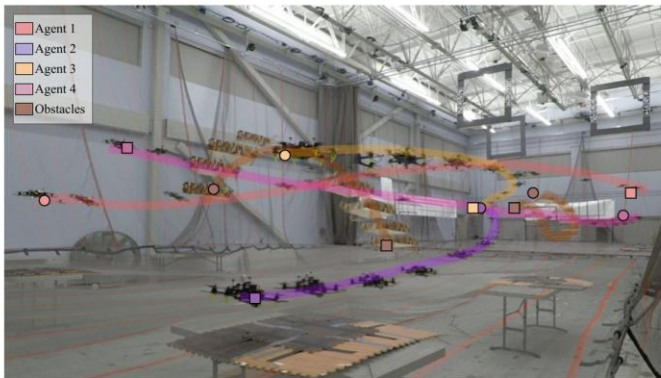
Mission Planning



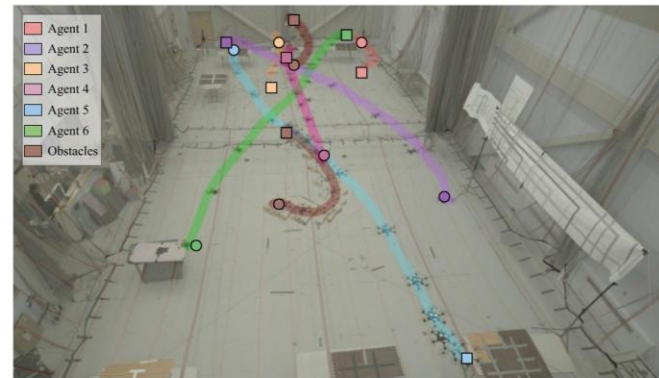
(a) 6 agents on mesh network.



(b) 2 agents with a dynamic obstacle.



(c) 4 agents with 2 dynamic obstacles.



(d) 6 agents with 2 dynamic obstacles.

Ref. Kondo, Kota & Figueroa, Reinaldo & Rached, Juan & Tordesillas, Jesús & Lusk, Parker & How, Jonathan. (2023). Robust MADER: Decentralized Multiagent Trajectory Planner Robust to Communication Delay in Dynamic Environments. 10.48550/arXiv.2303.06222.

Key Concepts in Multi-Agent UAV Control

Agent Characteristics

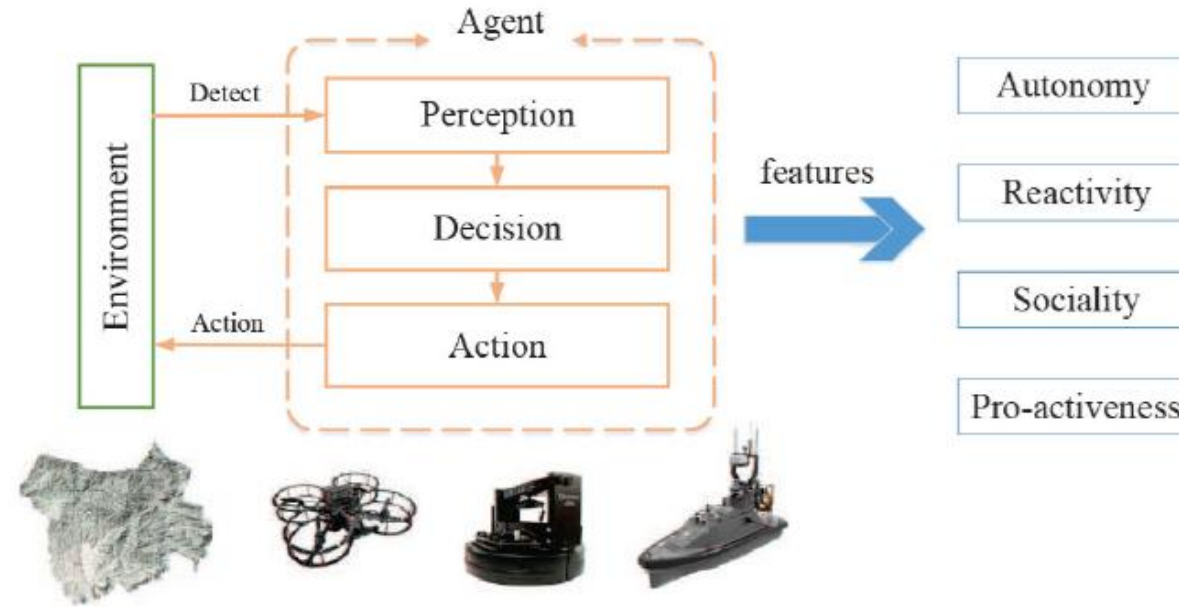
- **Autonomy:** Each agent can perform actions independently without needing external intervention.
- **Reactivity:** Agents can respond to changes in the environment.
- **Sociality:** Agents can communicate and cooperate with others.
- **Proactiveness:** Agents take actions to achieve goals, not just react to events.

Challenges:

1. **Limited Interaction Capabilities:** Agents have finite communication and sensing ranges.
2. **System Uncertainties:** Real-world systems include unknown disturbances, making coordination more difficult.
3. **Incomplete Information:** Agents often operate with only partial knowledge about the environment or other agents.

Key Concepts in Multi-Agent UAV Control

Agent & Features



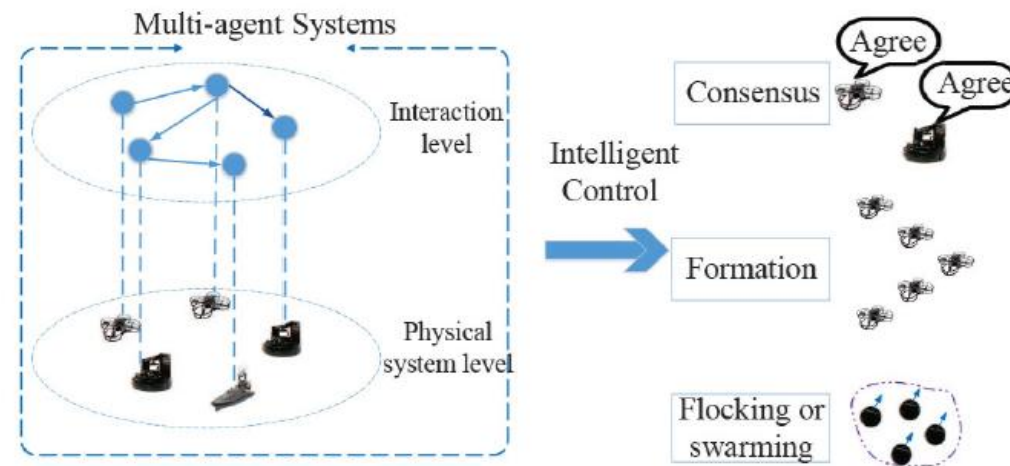
Agent and its features.

Ref. P. Shi and B. Yan, "A Survey on Intelligent Control for Multiagent Systems," in *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, vol. 51, no. 1, pp. 161-175, Jan. 2021, doi: 10.1109/TSMC.2020.3042823.

Intelligent Control in MAS

Approaches

- Consensus Problem: Agents must agree on a common value or course of action.
- Formation Control: Agents maintain a specific geometric arrangement (e.g., UAVs in formation).
- Flocking Control: Agents move cohesively while avoiding collisions and aligning direction.



Intelligent control for MASs.

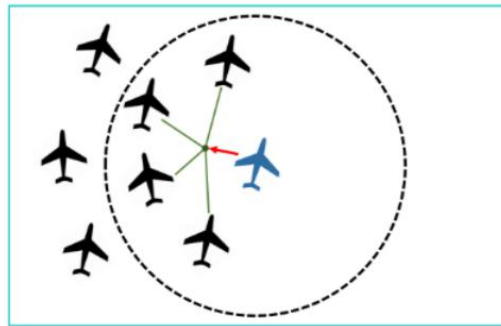
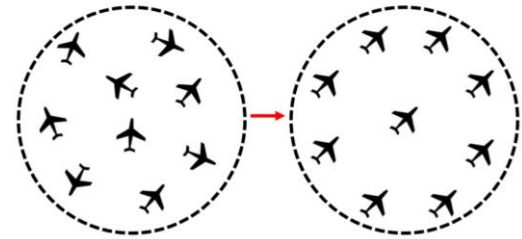
Ref. P. Shi and B. Yan, "A Survey on Intelligent Control for Multiagent Systems," in *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, vol. 51, no. 1, pp. 161-175, Jan. 2021, doi: 10.1109/TSMC.2020.3042823.

Intelligent Control in MAS

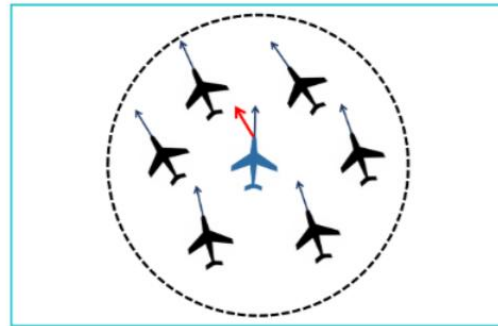
Reynolds' Rules for Flocking

- Reynolds' Rules:

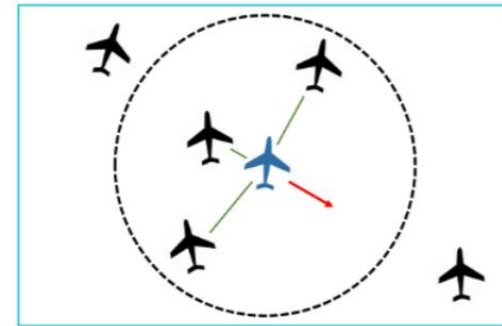
1. Alignment: align headings
2. Cohesion: steer towards average position of neighbors- towards c.g.
3. Separation: steer to maintain separation from neighbors



(a) cohesion



(b) alignment

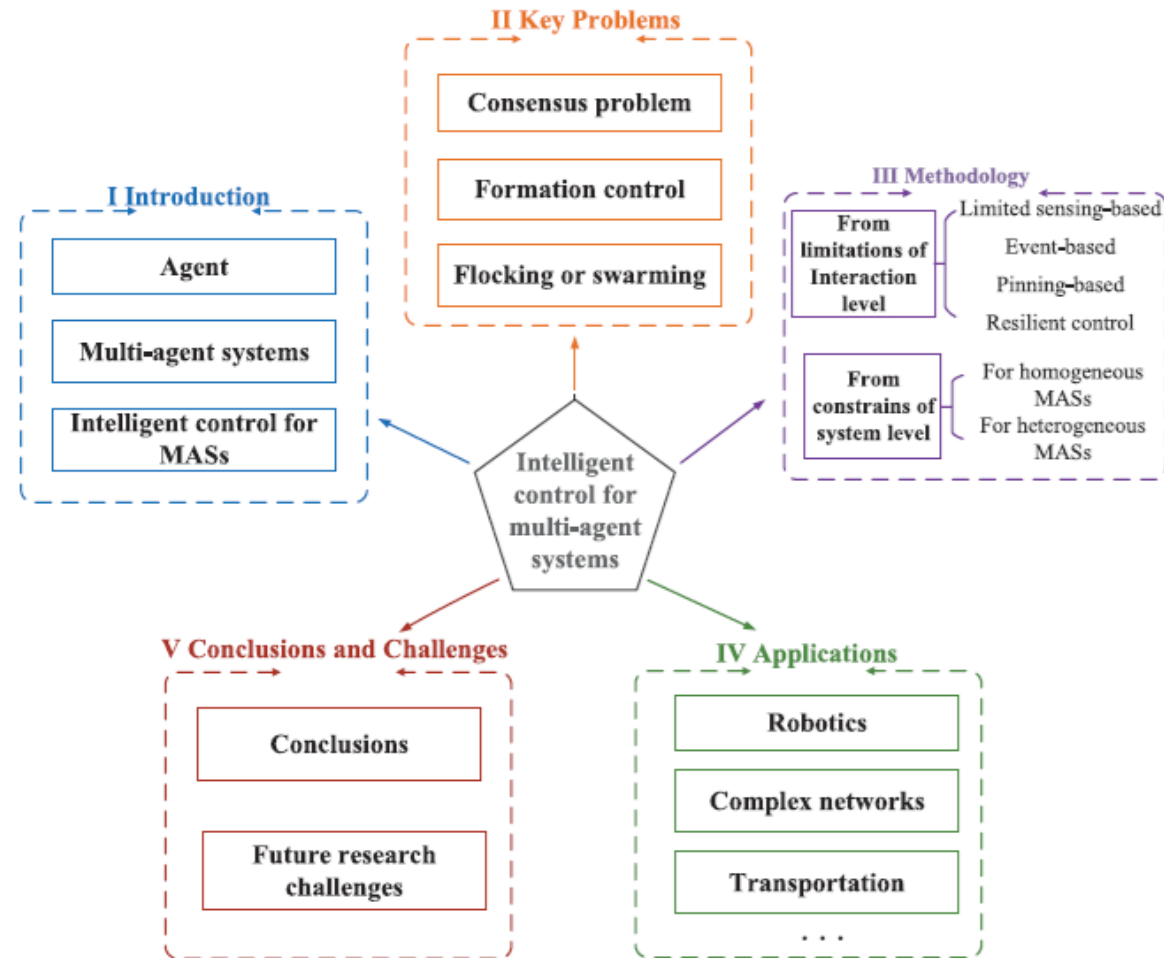


(c) separation

Ref: Song, Yeongho & Choi, Joonwon & Oh, Hyondong & Lee, Min & Lim, Seunghan & Lee, Jaemoon. (2019). Improvement of Decentralized Flocking Flight Efficiency of Fixed-Wing UAVs Using Inactive Agents.

Intelligent Control in MAS

Overall View

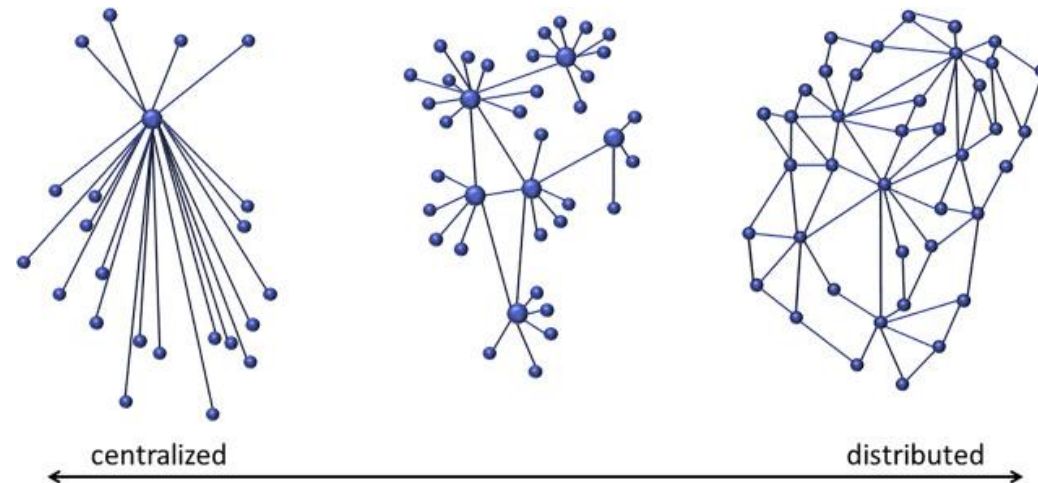


Ref. P. Shi and B. Yan, "A Survey on Intelligent Control for Multiagent Systems," in *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, vol. 51, no. 1, pp. 161-175, Jan. 2021, doi: 10.1109/TSMC.2020.3042823.

Intelligent Control in MAS

Centralized vs. Distributed Control

- **Centralized Control:** A central authority makes decisions for all agents. This can fail if the central node goes down.
- **Distributed Control:** Each agent makes decisions based on local information, making the system more robust and scalable.



Consensus Problem in MAS

Consensus

- Leaderless Consensus:
 - All agents aim to reach the same state (position, velocity, etc.) without a designated leader.
 - Example: Multiple UAVs aligning their heading direction to fly in unison.

Leader-Follower Consensus:

- One or more agents act as leaders, and the other agents follow their state or behavior.
- Example: A fleet of drones following a lead drone for search-and-rescue operations.

Consensus Problem in MAS

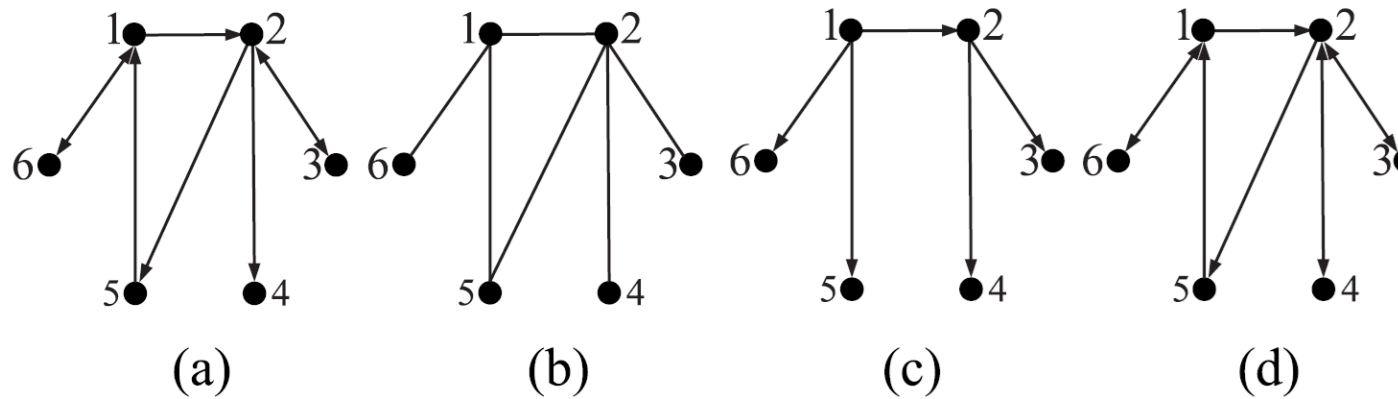
Linear Dynamics: Kronecker Product

- **Kronecker-Produkt**

- $A = [a_{ij}]$, B : $A \otimes B = [a_{ij}B]$
- $\|A \otimes I\| = \|A\|$
- $(A \otimes B)^T = A^T \otimes B^T$
- $A \otimes (B + C) = A \otimes B + A \otimes C$
- For scalar k , $(kA) \otimes B = A \otimes (kB) = k(A \otimes B)$
- For compatible matrices, $(A \otimes B)(C \otimes D) = AC \otimes BD$
- For square nonsingular matrices, $(A \otimes B)^{-1} = A^{-1} \otimes B^{-1}$
- Square matrix A and $A \otimes I$ have the same eigenvalues without considering multiplicity.

Consensus Problem in MAS

Graphs with Different Features



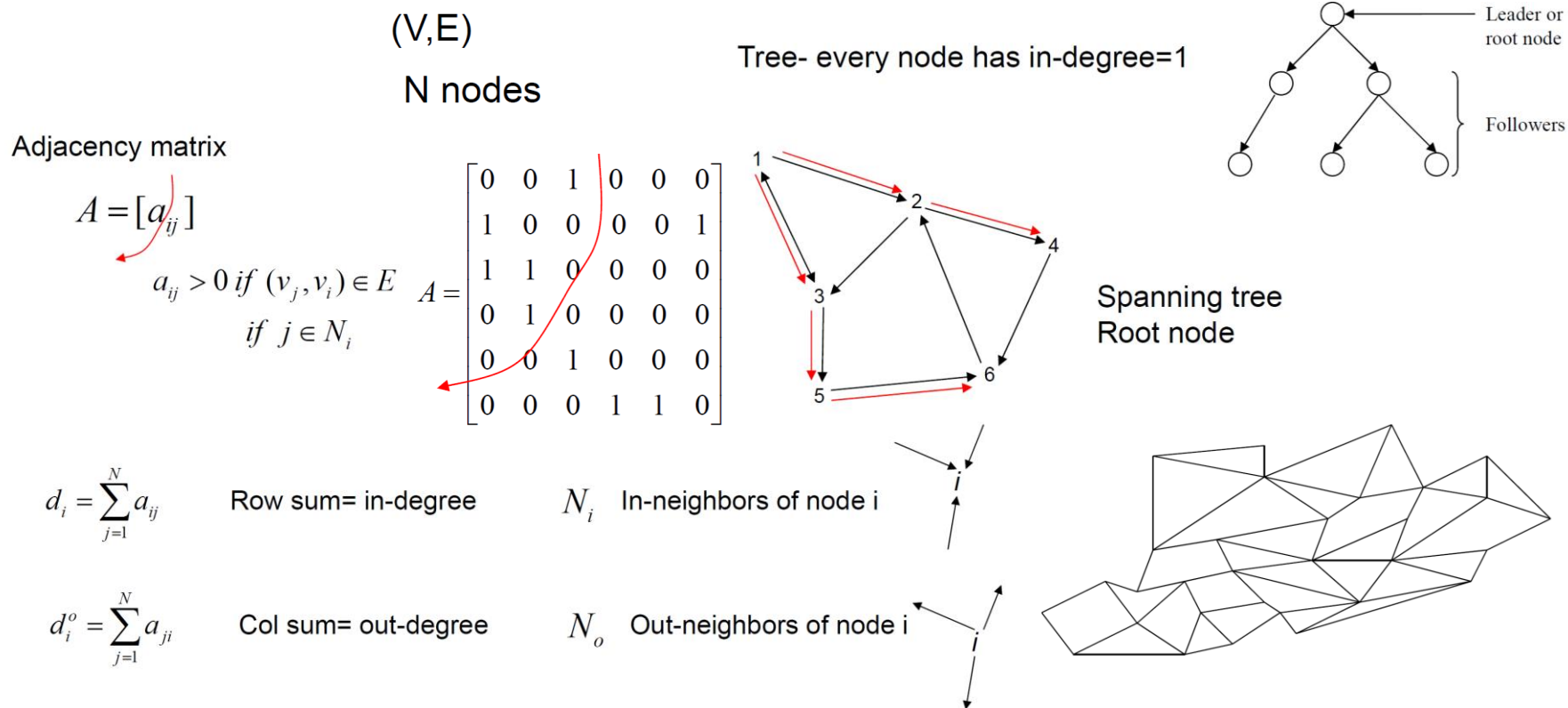
Graphs with different features. (a) Directed graph. (b) Undirected graph. (c) Spanning tree. (d) Connected graph.

Consensus Problem in MAS

Communication Graph

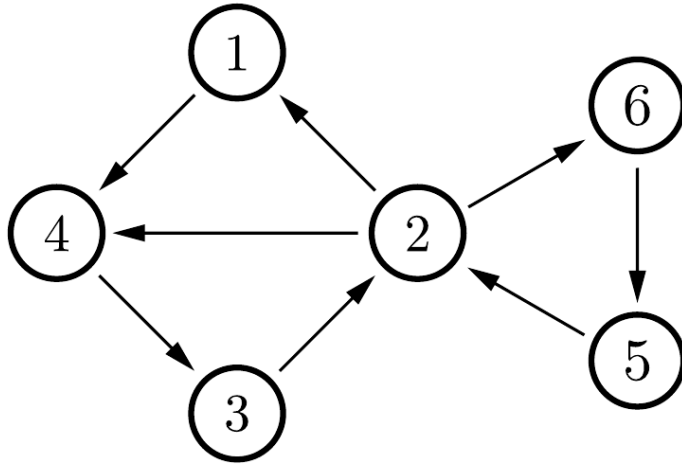
Strongly connected if for all nodes i and j there is a path from i to j .

Diameter: length of longest path between two nodes of the graph

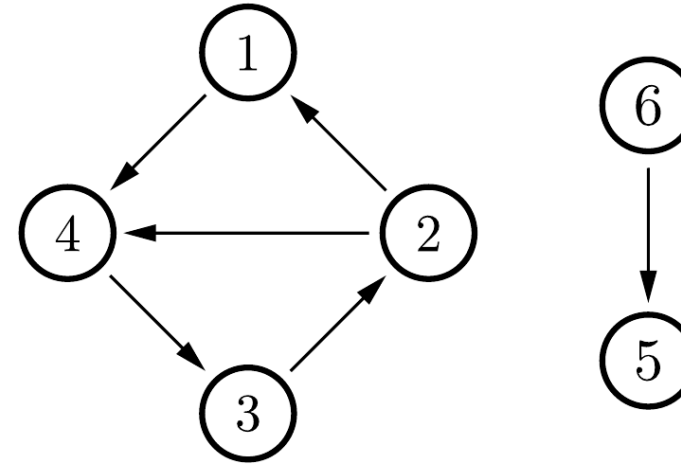


Consensus Problem in MAS

Consensus is possible if and only if one root can reach every agent



(a) Network in Example 3.2.



(b) Network in Example 3.3

Fig. 3.1 Networks with six nodes

Pass — Example 3.2: root 6 reaches all agents. **Fail — Example 3.3:** no root reaches both components.

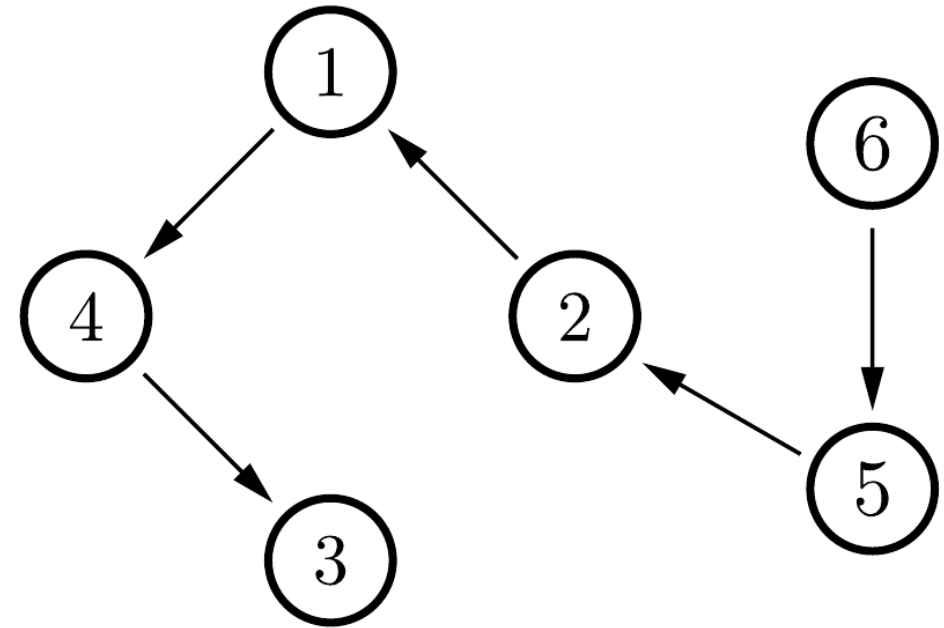
Directed path: a sequence of directed edges followed in the arrow direction.

Rooted spanning tree: a directed-tree subgraph containing all n agents, with one root that has a path to every other agent.

Consensus Problem in MAS

A spanning tree keeps only the edges needed for global reachability

Fig. 3.4 A spanning tree of the network in Fig. 3.1



Spanning = includes all n agents. **Tree** = the root has in-degree 0, every other agent has in-degree 1, and the root reaches each agent.

Minimum backbone: a tree with n agents has $n-1$ edges. Here, agent 6 is the root; extra links are not required for agreement.

Consensus Problem in MAS

The graph Laplacian converts local information flow into dynamics

From graph to matrices

$$A = [a_{ij}]$$

$a_{ij} \geq 0$ is the weight of the edge $j \rightarrow i$: agent i receives information from agent j .

$$d_i^{\text{in}} = \sum_j a_{ij}$$

d_i^{in} is agent i 's total incoming edge weight.

$$D_{\text{in}} = \text{diag}(d_1^{\text{in}}, \dots, d_n^{\text{in}})$$

D_{in} is the diagonal in-degree matrix.

$$L = D_{\text{in}} - A$$

L is the graph Laplacian; n is the number of agents.

Why the Laplacian matters

$$L \mathbf{1}_n = 0$$

$\mathbf{1}_n = [1, \dots, 1]^T$, so identical agent states form an equilibrium.

$\lambda = 0$ is simple $\Leftrightarrow$ G has a directed spanning tree

Simple means algebraic multiplicity one.

$\text{Re}(\lambda_k) > 0$ for every other eigenvalue λ_k

$\text{Re}(\lambda_k)$ is the real part; therefore $e^{(-Lt)}$ damps disagreement.

Collective model: with $x = [x_1, \dots, x_n]^T$, the dynamics are $\dot{x} = -Lx$.

Consensus Problem in MAS

A spanning tree gives agreement; balance gives the ordinary average

Balance test

$$d_i^{\text{in}} = \sum_j a_{ij} \quad \text{and} \quad d_i^{\text{out}} = \sum_j a_{ji}$$

d_i^{in} and d_i^{out} are the total incoming and outgoing edge weights at node i .

Balanced $\Leftrightarrow d_i^{\text{in}} = d_i^{\text{out}}$ for every i

Equivalent test: $\mathbf{1}_n^T \mathbf{L} = 0$

Every undirected graph is balanced.

Directed case: a graph may have a spanning tree without being balanced.

Consensus value

$$\mathbf{v}^T \mathbf{L} = 0, \quad \mathbf{v}^T \neq 0$$

$\mathbf{v}$ is a left eigenvector of $\mathbf{L}$ associated with eigenvalue 0.

$$\alpha = (\mathbf{v}^T \mathbf{x}(0)) / (\mathbf{v}^T \mathbf{1}_n)$$

α is the final common state; $\mathbf{x}(0)$ is the initial state vector.

Spanning tree + balanced: $\mathbf{v} \propto \mathbf{1}_n$, so $\alpha = (1/n) \sum_i x_i(0)$, the ordinary initial average.

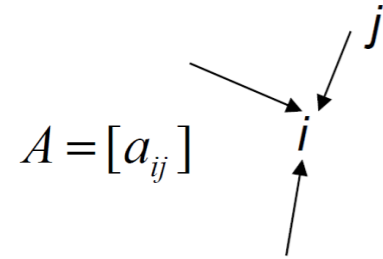
Spanning tree + unbalanced: consensus still occurs, but α is a weighted average.

Key distinction: the spanning tree gives agreement; balance selects the ordinary average.

Consensus Problem in MAS

Continuous-Time Protocol

Motivation Example: first order model



A simple first order system

$$\dot{x}_i = u_i$$

$$u_i = \sum_{j \in N_i} a_{ij} (x_j - x_i)$$

Classic local protocol

$$u_i = -x_i \sum_{j \in N_i} a_{ij} + \sum_{j \in N_i} a_{ij} x_j = -d_i x_i + [a_{i1} \quad \dots \quad a_{iN}] \begin{bmatrix} x_1 \\ \vdots \\ x_N \end{bmatrix}$$

$$u = \begin{bmatrix} u_1 \\ \vdots \\ u_N \end{bmatrix}$$

$$D = \begin{bmatrix} d_1 & & \\ & \ddots & \\ & & d_N \end{bmatrix}$$

x

$$u = -Dx + Ax = -(D - A)x = -Lx$$

$$\dot{x} = -Lx \quad \text{Closed-loop dynamics}$$

graph Laplacian matrix

Consensus Problem in MAS

Agent Dynamics

- **Linear dynamical system model**

Linear systems: (A, B) is stabilizable

$$\dot{x}_i(t) = Ax_i(t) + Bu_i(t)$$

Consensus algorithm

$$u_i(t) = K \sum_{j \in N_i} (x_j(t) - x_i(t))$$

Consensus Problem in MAS

Closed-Loop Dynamics

- **Augmented form**

Compact form

$$\dot{x}(t) = (I \otimes A)x(t) + (I \otimes B)u(t)$$

$$u(t) = -(L \otimes K)x(t)$$

Closed-loop system

$$\dot{x}(t) = [I \otimes A - (L \otimes BK)]x(t)$$

Solution

$$x(t) = e^{[I \otimes A - (L \otimes BK)]t}x(0)$$

Consensus Problem in MAS

State-Space Form

- **Matrix exponential**

State trajectory

$$x(t) = e^{[I \otimes A - (L \otimes BK)]t} x(0)$$

Matrix exponential

$$e^{[I \otimes A - L \otimes BK]t} = (P \otimes I) e^{[I \otimes A - J \otimes BK]t} (P^{-1} \otimes I)$$

Undirected graphs

$$I \otimes A - J \otimes BK = \text{diag}\{A, A - \lambda_2 BK, \dots, A - \lambda_N BK\}$$

Consensus Problem in MAS

Protocol

- **Stabilizing controller**

Find K such that all modes are simultaneously stabilized:

$$A - \lambda_i BK, \quad i = 2, \dots, N$$

Algebraic Riccati equation: (A, B) is stabilizable

$$PA + A^T P - \lambda_2 P B B^T P + Q = 0$$

Choose

$$K = \frac{1}{2} B^T P$$

Consensus Problem in MAS

Agreement

- **Consensus**

Trajectory

$$x(t) = (P \otimes I)e^{[I \otimes A - J \otimes BK]t}(P^{-1} \otimes I)x(0)$$

Consensus

$$x_i(t) \rightarrow e^{At}(q_1^T \otimes I)x(0)$$

$$q_1^T L = \mathbf{0}$$

$$I \otimes A - J \otimes BK = \text{diag}\{A, A - \lambda_2 BK, \dots, A - \lambda_N BK\}$$

Consensus Problem in MAS

Single and Double Integrator Dynamics

- **Special cases**

Double integrator dynamics

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

$$x_i(t) \rightarrow \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix} (q_1^T \otimes I) x(0)$$

Single integrator dynamics

$$A = 0$$

$$x_i(t) \rightarrow (q_1^T \otimes I) x(0)$$

Consensus Problem in MAS

Discrete consensus adds one design parameter: the step size

- **Agent-level update**

- $x_i[k+1] = x_i[k] + u_i[k]$
- $u_i[k] = \varepsilon \sum_{j \in N_i} (x_j[k] - x_i[k])$
- $i = 1, \dots, n$ identifies an agent; $k \in \{0, 1, \dots\}$ is the sample index.
- $x_i[k] \in \mathbb{R}$ is agent i 's state; $u_i[k] \in \mathbb{R}$ is its control input, added during one sample.
- N_i is the incoming-neighbor set; j indexes a neighbor that sends information to agent i .
- **Step size $\varepsilon > 0$:** scales the correction applied at each sample; larger ε means a stronger one-step update.

- **Collective form and step-size test**

- $x[k+1] = Px[k], \quad P = I_n - \varepsilon L$
- $x[k] = P^k x[0]$
- $\varepsilon \Delta < 1, \quad \Delta \triangleq \max_i d_i^{\text{in}}$
- $x[k] = [x_1[k], \dots, x_n[k]]^T$; $x[0]$ is the initial collective state, and P^k means k repeated updates.
- I_n is the $n \times n$ identity; L is the graph Laplacian; P is the Perron (update) matrix.
- $d_i^{\text{in}} = \sum_j a_{ij}$ is agent i 's total incoming edge weight, where a_{ij} weights $j \rightarrow i$; Δ is the maximum in-degree.
- **Convergence:** if $\varepsilon \Delta < 1$, consensus occurs exactly when the directed graph has a spanning tree.

Consensus Problem in MAS

The graph condition is shared; the convergence test changes

- **Continuous time**

- $\dot{x}(t) = -Lx(t), \quad t \geq 0$
- t is continuous time; $\dot{x}$ is the time derivative; $x(t) \in \mathbb{R}^n$ collects the n agent states; L is the graph Laplacian.
- **Graph test:** consensus $\Leftrightarrow$ the directed graph G has a spanning tree.
- $\alpha = (v^T x(0)) / (v^T 1_n), \quad v^T L = 0$
- α is the consensus value; v^T is a left zero-eigenvector of L ; 1_n is the all-ones vector.
- **continuous rate** = $\min_{i \geq 2} \text{Re}(\lambda_i(L))$
- $\lambda_i(L)$ are the nonzero Laplacian eigenvalues and Re is real part. A larger rate means faster decay; for connected undirected G , the rate is λ_2 .

- **Discrete time**

- $x[k+1] = Px[k], \quad P = I_n - \varepsilon L$
- k is the sample index; I_n is the identity matrix; $\varepsilon > 0$ is the step size; P is the update matrix.
- **Graph + step test:** if $\varepsilon \Delta < 1$, consensus $\Leftrightarrow G$ has a spanning tree; Δ is the maximum in-degree.
- $v^T P = v^T \Rightarrow$ the same α
- The left eigenvector v^T therefore assigns the same weights to the initial state as in continuous time.
- $\mu_i = 1 - \varepsilon \lambda_i(L), \quad |\mu_i| < 1$
- **discrete factor** = $\max_{i \geq 2} |\mu_i|$
- μ_i are disagreement eigenvalues of P ; $|\cdot|$ is complex modulus. A smaller factor means faster convergence.

Consensus Problem in MAS

Predict the consensus result before running the simulation

- **Graph and model checks**

- **1. Fix the arrow convention.** Use $a_{ij} > 0 \Leftrightarrow j \rightarrow i$; a_{ij} is the edge weight received by i from j .
- **2. Build the matrices.** $D_{in} = \text{diag}(\sum_j a_{ij})$, $L = D_{in} - A$. A is adjacency; $\text{diag}(\cdot)$ places values on the diagonal.
- **3. Check invariance.** Verify $L\mathbf{1}_n = \mathbf{0}$; $\mathbf{1}_n$ is the n -vector of ones.
- **4. Check reachability.** $\lambda = 0$ of L is simple $\Leftrightarrow G$ has a directed spanning tree. Simple means algebraic multiplicity one.
- **5. Check the target value.** For the ordinary average, verify $\mathbf{1}_n^T L = \mathbf{0}$. This is graph balance: each node has equal total incoming and outgoing weight.

- **Prediction and numerical checks**

- **1. Choose ϵ .** $\Delta = \max_i d_i^{in}$ and $\epsilon\Delta < 1$, where $d_i^{in} = \sum_j a_{ij}$ is agent i 's in-degree.
- **2. Form the update.** $P = I_n - \epsilon L$; verify that $P \geq 0$ entrywise and $P\mathbf{1}_n = \mathbf{1}_n$. Then P is row-stochastic; I_n is the identity.
- **3. Predict the limit.** Solve $\mathbf{v}^T L = \mathbf{0}$, normalize $\mathbf{v}^T \mathbf{1}_n = 1$, then $\alpha = \mathbf{v}^T \mathbf{x}[0]$. $\mathbf{x}[0]$ is the initial state vector; α is the predicted consensus.
- **4. Verify agreement.** Check $g[k] = \max_i x_i[k] - \min_i x_i[k] \rightarrow 0$; $g[k]$ is the max-min disagreement.
- **5. Verify the value.** The final state should approach $\alpha \mathbf{1}_n$. Explain any mismatch before trusting the plot.
- **predict $\rightarrow$ simulate $\rightarrow$ verify $\rightarrow$ explain**